Decision-Focused Learning via Tangent-Space Projection of Prediction Error

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Overview

1. **Regret Gradient via Tangent-Space Projection**

2. **Computing the Projected-Error Gradient**

Regret Gradient via Tangent-Space Projection

Problem Setup

Given contextual $\boldsymbol{x} \in \mathbb{R}^d$, predict cost vector $\hat{\boldsymbol{c}} = f_{\boldsymbol{\theta}}(\boldsymbol{x}) \in \mathbb{R}^n$ to estimate $\boldsymbol{c}$. Given $\hat{\boldsymbol{c}}$, consider convex problem:

$$\begin{aligned} \boldsymbol{z}^*(\hat{\boldsymbol{c}}) &\in \arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \{\phi(\boldsymbol{z}) + \hat{\boldsymbol{c}}^\top \boldsymbol{z}\} \\ \text{s.t.} \quad & \boldsymbol{A}_{p\times n}\boldsymbol{z} = \boldsymbol{b}, \quad \boldsymbol{G}_{m\times n}\boldsymbol{z} \le \boldsymbol{h}. \end{aligned} \tag{1}$$

DFL defines regret under real cost vector $\boldsymbol{c}$ as:

$$R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \underbrace{[\phi(\boldsymbol{z}^*(\hat{\boldsymbol{c}})) + \boldsymbol{c}^\top \boldsymbol{z}^*(\hat{\boldsymbol{c}})]}_{\text{predicted cost}} - \underbrace{[\phi(\boldsymbol{z}^*(\boldsymbol{c})) + \boldsymbol{c}^\top \boldsymbol{z}^*(\boldsymbol{c})]}_{\text{optimal cost}}. \tag{2}$$

Training $f_{\boldsymbol{\theta}}$ requires gradient $\nabla_{\boldsymbol{\theta}} R(\hat{\boldsymbol{c}}; \boldsymbol{c})$ by

$$\frac{\mathrm{d}R(\hat{\boldsymbol{c}}; \boldsymbol{c})}{\mathrm{d}\boldsymbol{\theta}} = \frac{\partial R}{\partial \boldsymbol{z}^*} \frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}} \frac{\partial \hat{\boldsymbol{c}}}{\partial \boldsymbol{\theta}}. \tag{3}$$

And the gradient w.r.t. $\hat{\boldsymbol{c}}$ is

$$\frac{\partial R}{\partial \hat{\boldsymbol{c}}} = \left(\frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}}\right)^\top (\nabla_{\boldsymbol{z}^*} \phi(\boldsymbol{z}^*) + \boldsymbol{c}). \tag{4}$$

Active Set and Local Regularity

Main problem: active set $(\mathcal{A}(\hat{\boldsymbol{c}}) := \{i \in [m] : \boldsymbol{g}_i^\top \boldsymbol{z}^*(\hat{\boldsymbol{c}}) = h_i\})$ may change and thus $\hat{\boldsymbol{c}} \mapsto \boldsymbol{z}^*(\hat{\boldsymbol{c}})$ may be non-smooth.

General framework (notations not consistent)

Consider $\boldsymbol{y}^*(\boldsymbol{x}) = \arg\min_{\boldsymbol{y} \in \mathcal{Y}} g(\boldsymbol{x}, \boldsymbol{y})$, $\boldsymbol{x}$ may be viewed as a parameter.

- If $\mathcal{Y} = \mathbb{R}^n$ and g is smooth and convex, then optimal solution $\boldsymbol{y}^*(\boldsymbol{x})$ is given by $\nabla_{\boldsymbol{y}} g(\boldsymbol{x}, \boldsymbol{y}^*(\boldsymbol{x})) = 0$. Then, by implicit function theorem, $\boldsymbol{y}^*(\boldsymbol{x})$ is smooth when $\nabla^2_{\boldsymbol{yy}} g(\boldsymbol{x}, \boldsymbol{y}^*(\boldsymbol{x}))$ is non-singular.
- But if constrained (e.g. $\mathcal{Y} = \{\boldsymbol{y} : \boldsymbol{G}\boldsymbol{y} \leq \boldsymbol{h}\}$), then the optimality condition becoems $0 \in \nabla_{\boldsymbol{y}} g(\boldsymbol{x}, \boldsymbol{y}^*(\boldsymbol{x})) + \mathcal{N}_{\mathcal{Y}}(\boldsymbol{y}^*(\boldsymbol{x}))$, where $\mathcal{N}_{\mathcal{Y}}(\boldsymbol{y})$ is the normal cone of $\mathcal{Y}$ at $\boldsymbol{y}$.
- Therefore, when $\boldsymbol{x}$ changes, the optimal solution $\boldsymbol{y}^*(\boldsymbol{x})$ may change, shifting from one face of $\mathcal{Y}$ to another, and thus the active set may change. In short, $\boldsymbol{x} \mapsto \mathcal{A}(\boldsymbol{x}) \mapsto \boldsymbol{y}^*(\boldsymbol{x})$, and $\mathcal{A}(\boldsymbol{x})$ may be discrete.
- Yet within each active set, $\boldsymbol{y}^*(\boldsymbol{x})$ is smooth, thus it is piecewise smooth.

Appendix: Normal Cone I

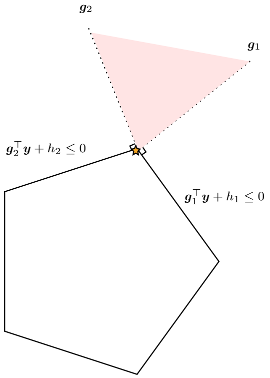
Definition (Normal Cone). For a closed convex set $\mathcal{Y} \subseteq \mathbb{R}^n$ and $\boldsymbol{y} \in \mathcal{Y}$, the normal cone of $\mathcal{Y}$ at $\boldsymbol{y}$ is defined as $\mathcal{N}_{\mathcal{Y}}(\boldsymbol{y}) := \{\boldsymbol{v} \in \mathbb{R}^n : \boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y}) \leq 0,\ \forall \boldsymbol{y}' \in \mathcal{Y}\}$.

- Tl;dr: vectors that point outward of $\mathcal{Y}$ at ref $\boldsymbol{y}$.
- For polyhedral set $\mathcal{Y} = \{\boldsymbol{y} : \boldsymbol{G}\boldsymbol{y} \leq \boldsymbol{h}\}$, the normal cone is

$$\mathcal{N}_{\mathcal{Y}}(\boldsymbol{y}) = \left\{ \sum_{i \in \mathcal{A}(\boldsymbol{y})} \lambda_i \boldsymbol{g}_i : \lambda_i \geq 0 \right\} := \text{cone}\left\{\boldsymbol{g}_i : i \in \mathcal{A}(\boldsymbol{y})\right\}$$

  where $\mathcal{A}(\boldsymbol{y}) := \{i : \boldsymbol{g}_i^\top \boldsymbol{y} = h_i\}$ is the active set at $\boldsymbol{y}$, $\boldsymbol{g}_i^\top$ is the i-th row of $\boldsymbol{G}$, and cone$\{S\}$ is the conic hull of S.

Thus, for such constrained problem, the optimality condition is $-\nabla_{\boldsymbol{y}} g(\boldsymbol{x}, \boldsymbol{y}^*(\boldsymbol{x})) \in \mathcal{N}_{\mathcal{Y}}(\boldsymbol{y}^*(\boldsymbol{x}))$, i.e., the descent motion can be blocked by the the combination of active constraints (*which is exactly the more general form of KKT condition*).



Appendix: Normal Cone II

In a more convex analysis perspective, consider extended value indicator $\delta_{\mathcal{Y}}(\boldsymbol{y}) = 0$ if $\boldsymbol{y} \in \mathcal{Y}$, $+\infty$ otherwise. Then $\min_{\boldsymbol{y}\in\mathcal{Y}} g(\boldsymbol{y}) = \min_{\boldsymbol{y}\in\mathbb{R}^n}\{g(\boldsymbol{y}) + \delta_{\mathcal{Y}}(\boldsymbol{y})\} := \min_{\boldsymbol{y}\in\mathbb{R}^n} F(\boldsymbol{y})$.

Fact: $\partial\delta_{\mathcal{Y}}(\boldsymbol{y}) = \mathcal{N}_{\mathcal{Y}}(\boldsymbol{y})$

Proof. By subdifferential $\partial\delta_{\mathcal{Y}}(\boldsymbol{y}) = \{\boldsymbol{v} : \delta_{\mathcal{Y}}(\boldsymbol{y}') \geq \delta_{\mathcal{Y}}(\boldsymbol{y}) + \boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y}),\ \forall \boldsymbol{y}'\}$.

Case	$\delta_{\mathcal{Y}}(\boldsymbol{y})$	$\delta_{\mathcal{Y}}(\boldsymbol{y}')$	Subgradient inequality	Consequence
$\boldsymbol{y} \in \mathcal{Y},\ \boldsymbol{y}' \in \mathcal{Y}$	0	0	$0 \geq \boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y})$	$\boxed{\boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y}) \leq 0}$
$\boldsymbol{y} \in \mathcal{Y},\ \boldsymbol{y}' \notin \mathcal{Y}$	0	$+\infty$	$+\infty \geq \boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y})$	Automatically satisfied
$\boldsymbol{y} \notin \mathcal{Y},\ \boldsymbol{y}' \in \mathcal{Y}$	$+\infty$	0	$0 \geq +\infty$	Impossible
$\boldsymbol{y} \notin \mathcal{Y},\ \boldsymbol{y}' \notin \mathcal{Y}$	$+\infty$	$+\infty$	$+\infty \geq +\infty$	No additional restriction

And the only non-trivial case is exactly the definition of normal cone.

Thus, the optimality condition can be written as $0 \in \nabla_{\boldsymbol{y}} g(\boldsymbol{y}^*) + \partial\delta_{\mathcal{Y}}(\boldsymbol{y}^*) = \partial F(\boldsymbol{y}^*)$.

Active set and Local Regularity (cont.) I

Back to the original problem.

Given $\hat{\boldsymbol{c}}$, let $\mathcal{A}(\hat{\boldsymbol{c}}) := \{i : (\boldsymbol{G}\boldsymbol{z}^*(\hat{\boldsymbol{c}}))_i = h_i\}$ be the active set at $\boldsymbol{z}^*(\hat{\boldsymbol{c}})$. Define the substack of the active constraint matrix

$$\boldsymbol{J}(\hat{\boldsymbol{c}}) := \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{G}_{\mathcal{A}(\hat{\boldsymbol{c}})} \end{bmatrix} \in \mathbb{R}^{k \times n}, \quad k = p + |\mathcal{A}(\hat{\boldsymbol{c}})|.$$

This paper focuses on each neighbourhood, where $\mathcal{A}(\hat{\boldsymbol{c}})$ is unchanged, i.e., $\mathcal{A}(\hat{\boldsymbol{c}}) = \mathcal{A}(\hat{\boldsymbol{c}} + d\hat{\boldsymbol{c}})$ for sufficiently small perturbation $d\hat{\boldsymbol{c}}$. Thus, $\boldsymbol{J}(\hat{\boldsymbol{c}}) = \boldsymbol{J}(\hat{\boldsymbol{c}} + d\hat{\boldsymbol{c}}) := \boldsymbol{J}$ is fixed, and the solution map $\boldsymbol{z}^*(\hat{\boldsymbol{c}})$ is smooth w.r.t. $\hat{\boldsymbol{c}}$.

Note on $\boldsymbol{J}$

- Since all constraints are linear, $\boldsymbol{J}(\hat{\boldsymbol{c}})$ is also the Jacobian of the active constraints at $\boldsymbol{z}^*(\hat{\boldsymbol{c}})$.
- $\boldsymbol{J} : \mathbb{R}^n \to \mathbb{R}^k$ is a linear map from decision space to active constraint space.
- $\boldsymbol{J}d\boldsymbol{z}$ can be viewed as: if $\boldsymbol{z}$ has a tiny perturbation $d\boldsymbol{z}$, then $\boldsymbol{J}d\boldsymbol{z}$ is the corresponding change of the active constraint values on each row.

Active set and Local Regularity (cont.) II

To further ensure smothness within each region, it requires the following classical regularity conditions:

- $\boldsymbol{H} = \nabla^2 \phi(\boldsymbol{z}^*(\hat{\boldsymbol{c}}))$ is positive definite.
- LICQ (linearly independent constraint qualification): $\boldsymbol{J}(\hat{\boldsymbol{c}})$ is full row rank.
- Strict complementarity: For active inequality constraints $i \in \mathcal{A}$, the corresponding Lagrange multiplier $\mu_i^* > 0$; for inactive constraints $i \notin \mathcal{A}$, $\mu_i^* = 0$.

These assumptions ensure: solution map is locally unique, continuous, and active set is invariant under sufficiently small perturbation of $\hat{\boldsymbol{c}}$.

Sensitivity via a Reduced KKT System I

The origin KKT system is

$$\begin{aligned} \nabla\phi(\boldsymbol{z}^*) + \boldsymbol{A}^\top \boldsymbol{\lambda}^* + \boldsymbol{G}^\top \boldsymbol{\mu}^* + \hat{\boldsymbol{c}} &= 0 \\ \boldsymbol{A}\boldsymbol{z}^* = \boldsymbol{b},\ \boldsymbol{G}\boldsymbol{z}^* \leq \boldsymbol{h}, \boldsymbol{\mu}^* &\geq 0, \\ \mathrm{diag}(\boldsymbol{\mu}^*)(\boldsymbol{G}\boldsymbol{z}^* - \boldsymbol{h}) &= 0. \end{aligned} \tag{5}$$

Under the regularity conditions, focusing on local neighbourhood of $\hat{\boldsymbol{c}}$, the reduced KKT system is

$$\nabla\phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top \boldsymbol{y}^* = 0,\ \boldsymbol{J}\boldsymbol{z}^* = \tilde{\boldsymbol{b}}, \tag{6}$$

where

$$\boldsymbol{J} = \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{G}_{\mathcal{A}(\hat{\boldsymbol{c}})} \end{bmatrix}, \quad \boldsymbol{y}^* = \begin{bmatrix} \boldsymbol{\lambda}^* \\ \boldsymbol{\mu}^*_{\mathcal{A}(\hat{\boldsymbol{c}})} \end{bmatrix}, \quad \tilde{\boldsymbol{b}} = \begin{bmatrix} \boldsymbol{b} \\ \boldsymbol{h}_{\mathcal{A}(\hat{\boldsymbol{c}})} \end{bmatrix}. \tag{7}$$

Form equation 6 into a linear system:

$$F(\boldsymbol{z}, \boldsymbol{y}, \hat{\boldsymbol{c}}) = \begin{bmatrix} \nabla\phi(\boldsymbol{z}) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top \boldsymbol{y} \\ \boldsymbol{J}\boldsymbol{z} - \tilde{\boldsymbol{b}} \end{bmatrix} = 0. \tag{8}$$

Sensitivity via a Reduced KKT System II

Taking the differential of F w.r.t. $(\boldsymbol{z}, \boldsymbol{y})$ and $\hat{\boldsymbol{c}}$ gives the sensitivity of the solution w.r.t. $\hat{\boldsymbol{c}}$:

$$\boxed{\begin{bmatrix} \boldsymbol{H} & \boldsymbol{J}^\top \\ \boldsymbol{J} & 0 \end{bmatrix} \begin{bmatrix} d\boldsymbol{z} \\ d\boldsymbol{y} \end{bmatrix} = \begin{bmatrix} -d\hat{\boldsymbol{c}} \\ 0 \end{bmatrix}} \qquad \text{where } \boldsymbol{H} = \nabla^2 \phi(\boldsymbol{z}^*). \tag{9}$$

Now consider its geometric interpretation.
This equation is invertivle under the regularity conditions, and thus the first row gives $\boldsymbol{H} d\boldsymbol{z} + \boldsymbol{J}^\top d\boldsymbol{y} = -d\hat{\boldsymbol{c}}$.

$$\implies d\boldsymbol{z} = -\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top d\boldsymbol{y}. \tag{10}$$

Plug this in $\boldsymbol{J} d\boldsymbol{z} = 0$ gives $\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top d\boldsymbol{y} = -\boldsymbol{J}\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$, and thus $d\boldsymbol{y} = -(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$.
Finally, plug $d\boldsymbol{y}$ back to equation 10 gives

$$d\boldsymbol{z} = -\left(\boldsymbol{H}^{-1} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{H}^{-1}\right) d\hat{\boldsymbol{c}} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}}. \tag{11}$$

$$\boxed{\frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}} = -\boldsymbol{P}_H} \tag{12}$$

Sensitivity via a Reduced KKT System III

In last page, we denote $\boldsymbol{P}_H = \boldsymbol{H}^{-1} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{H}^{-1} := \boldsymbol{\Pi}_H\boldsymbol{H}^{-1}$, where

$$\boxed{\boldsymbol{\Pi}_H := \boldsymbol{I} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}} \tag{13}$$

Mathematically, $\boldsymbol{\Pi}_H$ is a projection matrix onto the kernel of $\boldsymbol{J}$ ($\ker(\boldsymbol{J}) := \{\boldsymbol{v} : \boldsymbol{J}\boldsymbol{v} = 0\}$), under $\boldsymbol{H}$-orthogonal (i.e., by conisdering the inner product induced by $\langle \boldsymbol{u}, \boldsymbol{v}\rangle_{\boldsymbol{H}} = \boldsymbol{u}^\top\boldsymbol{H}\boldsymbol{v}$).

Sensitivity via a Reduced KKT System IV

Understanding of $\boldsymbol{J}$, its kernel and range

- Kernel of $\boldsymbol{J}$: $\ker(\boldsymbol{J}) = \{\boldsymbol{v} : \boldsymbol{J}\boldsymbol{v} = 0\} \subseteq \mathbb{R}^n$, i.e., all directions that do not change the active constraint values. This is exactly the meaning of the tangent space, and denote $\mathcal{T}_{\boldsymbol{z}^*} := \ker(\boldsymbol{J})$.
- Range of $\boldsymbol{J}$: $\mathrm{range}(\boldsymbol{J}) = \{\boldsymbol{u} : \boldsymbol{u} = \boldsymbol{J}\boldsymbol{v},\ \boldsymbol{v} \in \mathbb{R}^n\} \subseteq \mathbb{R}^k$, i.e., all possible changing values of the active constraints from decision perturbations. (But we don't really care).
- Range of $\boldsymbol{J}^\top (\mathbb{R}^k \to \mathbb{R}^n)$: Recall the KKT equation 6 $\nabla_{\boldsymbol{z}^*}\phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top \boldsymbol{y}^* = 0$, $\boldsymbol{y}^*$ is the Lagrange multiplier (as weights of active constraints' gradients), to balance the descent direction $-\nabla_{\boldsymbol{z}^*}\phi(\boldsymbol{z}^*) - \hat{\boldsymbol{c}}$.

  Thus, if for arbitrary $\boldsymbol{y} \in \mathbb{R}^k$, $\boldsymbol{J}^\top \boldsymbol{y}$ is the linear combination of active constraints' gradients, i.e., the normal space of the tangent space. Denote $\mathcal{N}_{\boldsymbol{z}^*} := \mathrm{range}(\boldsymbol{J}^\top)$.

  As a matter of fact:

$$\mathrm{range}(\boldsymbol{J}^\top) = \ker(\boldsymbol{J})^\perp = \mathcal{T}_{\boldsymbol{z}^*}^\perp = \mathcal{N}_{\boldsymbol{z}^*}. \tag{14}$$

Sensitivity via a Reduced KKT System V

Recall $\boldsymbol{P}_H = \boldsymbol{H}^{-1} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{H}^{-1} := \boldsymbol{\Pi}_H\boldsymbol{H}^{-1}$.

Then revisit this $\boldsymbol{\Pi}_H$: *projection onto* $\ker(\boldsymbol{J})$ *under* $\boldsymbol{H}$*-orthogonal*. It means that, for any direction $\boldsymbol{v} \in \mathbb{R}^n$, $\boldsymbol{\Pi}_H\boldsymbol{v}$ is the closest vector to $\boldsymbol{v}$ in $\ker(\boldsymbol{J})$ under the $\boldsymbol{H}$-norm, and it will always be in the tangent space, not changing the active constraints' values.

Sensitivity via a Reduced KKT System VI

Recall that equation 11: $d\boldsymbol{z} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}} = -\boldsymbol{\Pi}_H \boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$ (i.e.,$\boldsymbol{P}_H : d\hat{\boldsymbol{c}} \mapsto d\boldsymbol{z}$). Then, the geometric interpretation of $\boldsymbol{P}_H$ is:

- $\boldsymbol{\Pi}_H$ is the projection of movement. Yet $d\hat{\boldsymbol{c}}$ is the perturbation of the predicted cost, not the decision.
- If this problem is unconstrained, i.e., $\min_{\boldsymbol{z}} \phi(\boldsymbol{z}) + \hat{\boldsymbol{c}}^\top \boldsymbol{z}$, then $\nabla \phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} = 0$, and thus $d\boldsymbol{z} = -\nabla^2 \phi(\boldsymbol{z}^*)^{-1} d\hat{\boldsymbol{c}} = -\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$. So first we know: $\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$ maps the perturbation of predicted cost to the perturbation of decision.
- Then, given that there are active constraints, $\boldsymbol{\Pi}_H$ projects the perturbation of decision onto the tangent space, to ensure that the active constraints' values are unchanged.

Sensitivity via a Reduced KKT System VII

Then the paper shows that,

$$\boldsymbol{J}\boldsymbol{P}_H = 0, \qquad \boldsymbol{P}_H \boldsymbol{J}^\top = 0. \tag{15}$$

- The first one is obvious, since $\boldsymbol{P}_H$ projects onto $\ker(\boldsymbol{J})$, thus $\boldsymbol{J}\boldsymbol{P}_H = 0$. *Any primal movement caused by the perturbation of predicted cost will not change the active constraints' values (in first-order approximation).*
- The second is less obvious.
  Consider a special cost perturbation $d\hat{\boldsymbol{c}} \in \mathrm{range}(\boldsymbol{J}^\top)$, i.e., $d\hat{\boldsymbol{c}} = \boldsymbol{J}^\top d\boldsymbol{y}$ for some $d\boldsymbol{y}$. Then the KKT (equation 6) gives $\nabla\phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top(\boldsymbol{y}^* + d\boldsymbol{y}) = 0$, thus the optimal solution does not change; we only need to adjust the Lagrange multiplier $\boldsymbol{y}^*$ to balance the new cost perturbation. Thus, $d\boldsymbol{z} = 0$.
  Meanwhile, equation 11 gives $d\boldsymbol{z} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}} = -\boldsymbol{P}_H \boldsymbol{J}^\top d\boldsymbol{y} = 0$, thus $\boldsymbol{P}_H \boldsymbol{J}^\top = 0$.
  Any cost perturbation in the normal space of the tangent space does not change the optimal solution.

Regret Gradient and the Projected Prediction Error I

Recall that, our key is still to solve $\frac{\partial R}{\partial \hat{\boldsymbol{c}}} = \left(\frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}}\right)^\top (\nabla_{\boldsymbol{z}^*}\phi(\boldsymbol{z}^*) + \boldsymbol{c})$.

- The first term, $\frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}} = -\boldsymbol{P}_H$ (equation 12) measures how much decision $\boldsymbol{z}^*$ changes w.r.t. predicted cost $\hat{\boldsymbol{c}}$. Moreover, note that $\boldsymbol{P}_H = \boldsymbol{P}_H^\top$ is symmetric.
- The second term, from reduced KKT (equation 6), $\nabla_{\boldsymbol{z}^*}\phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top \boldsymbol{y}^* = 0$, thus $\nabla_{\boldsymbol{z}^*}\phi(\boldsymbol{z}^*) + \boldsymbol{c} = (\boldsymbol{c} - \hat{\boldsymbol{c}}) - \boldsymbol{J}^\top \boldsymbol{y}^*$.

Plug in the two gives

$$\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = -\boldsymbol{P}_H(\boldsymbol{c} - \hat{\boldsymbol{c}} - \boldsymbol{J}^\top \boldsymbol{y}^*) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}) + \underbrace{\boldsymbol{P}_H \boldsymbol{J}^\top}_{=0} \boldsymbol{y}^* = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}). \tag{16}$$

Denote $\boldsymbol{e} = \hat{\boldsymbol{c}} - \boldsymbol{c}$ as the prediction error, then the key formula, i.e. projected prediction error, is

$$\boxed{\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H \boldsymbol{e}} \tag{17}$$

Compared with MSE gradient $\nabla_{\hat{\boldsymbol{c}}} \mathcal{L}_{\mathsf{MSE}} = \hat{\boldsymbol{c}} - \boldsymbol{c} = \boldsymbol{e}$, it implies: *each direction of pred.error needs to be adjusted*.

Regret Gradient and the Projected Prediction Error II

Corollary (Normal Filtering and Normal Invariance)

Let $\mathcal{N} = \mathrm{range}(\boldsymbol{J}^\top)$ *be the normal space of the tangent space* $\mathcal{T}$*. For any* $\boldsymbol{\delta}_N \in \mathcal{N}$*,*

$$\boldsymbol{P}_H \boldsymbol{\delta}_N = 0. \tag{18}$$

If active set is unchanged, then for all $\alpha \in \mathbb{R}$*,*

$$\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}} + \alpha \boldsymbol{\delta}_N; \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} + \alpha \boldsymbol{\delta}_N - \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}) + \alpha \underbrace{\boldsymbol{P}_H \boldsymbol{\delta}_N}_{=0} = \nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}). \tag{19}$$

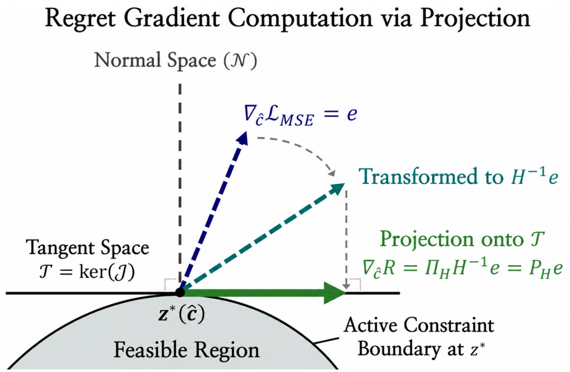
- Here, equation 18 is basically the same as $\boldsymbol{P}_H \boldsymbol{J}^\top = 0$.
- Then, the corollary shows that, since $\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c})$, then if there is anything normal in the prediction error: $\boldsymbol{e} = \boldsymbol{e}_T + \boldsymbol{\delta}_N$, then all those normal components $\boldsymbol{\delta}_N$ will be filtered out, contributing nothing to the regret gradient.

Thus, the regret gradient is invariant to any normal perturbation of the predicted cost $\hat{c}$, as long as the active set is unchanged.

Regret Gradient and the Projected Prediction Error III

Finally, the key takeaway is this decomposition of the regret gradient:

$$\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H \boldsymbol{e} = \boldsymbol{\Pi}_H \boldsymbol{H}^{-1} \boldsymbol{e} = \underbrace{\left(\boldsymbol{I} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top (\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\right)}_{\text{projection onto tangent space}} \underbrace{\boldsymbol{H}^{-1}\boldsymbol{e}}_{\text{preconditioning by Hessian}} . \tag{20}$$



Computing the Projected-Error Gradient

General Framework of PEAR

From last section,

$$\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}).$$

The general update is still the classic first-order gradient descent:

$$\begin{aligned} \frac{dR}{d\boldsymbol{\theta}} &= \frac{\partial \hat{\boldsymbol{c}}}{\partial \boldsymbol{\theta}}^{\top} \nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) \\ &= \left(\frac{\partial \hat{\boldsymbol{c}}}{\partial \boldsymbol{\theta}}\right)^{\top} \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}). \end{aligned} \tag{21}$$

The key is to compute

$$\boldsymbol{g} := \nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c})$$

efficiently, rather than forming $\boldsymbol{P}_H \in \mathbb{R}^{n \times n}$ explicitly.

Algorithm 1 PEAR: Projected Error as Regret-gradient

Input: Predicted costs $\hat{\boldsymbol{c}} = f_{\boldsymbol{\theta}}(\boldsymbol{x})$, true costs $\boldsymbol{c}$, problem data $(\boldsymbol{A}, \boldsymbol{b}, \boldsymbol{G}, \boldsymbol{h})$
Input: Hessian $\boldsymbol{H} = \nabla^2 \phi(\boldsymbol{z}^*)$ (or $\boldsymbol{H} = \lambda \boldsymbol{I}$ for LP), injection strength $\beta \in [0, 1]$
Output: Gradient signal $\boldsymbol{g} = \nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c})$

// **Forward: Active-Set Detection**
Solve (1) with cost $\hat{\boldsymbol{c}}$ to obtain primal-dual pair $(\boldsymbol{z}^*, \boldsymbol{y}^*)$
Identify active set $\mathcal{A}$ via complementary slackness (16)
Form active constraint Jacobian $\boldsymbol{J} \leftarrow \left[\boldsymbol{A}^{\top}, \boldsymbol{G}_{\mathcal{A}}^{\top}\right]^{\top}$
// **Backward: Compute $\boldsymbol{P}_H \boldsymbol{e}$ via Reduced System**
Compute prediction error $\boldsymbol{e} \leftarrow \hat{\boldsymbol{c}} - \boldsymbol{c}$
Compute $\boldsymbol{x} \leftarrow \boldsymbol{H}^{-1}\boldsymbol{e}$ and $\boldsymbol{r} \leftarrow \boldsymbol{J}\boldsymbol{x}$
Solve $(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^{\top})\boldsymbol{v} = \boldsymbol{r}$ for $\boldsymbol{v}$
Compute projected gradient $\boldsymbol{g} \leftarrow \boldsymbol{x} - \boldsymbol{H}^{-1}\boldsymbol{J}^{\top}\boldsymbol{v}$
// **Optional: Normal injection for LP**
if $\beta > 0$ **then**
  Compute normal component $\boldsymbol{n} \leftarrow \boldsymbol{H}^{-1}\boldsymbol{J}^{\top}\boldsymbol{v}$
  $\boldsymbol{g} \leftarrow \boldsymbol{g} + \beta \cdot \frac{\|\boldsymbol{g}\|}{\|\boldsymbol{n}\|} \cdot \boldsymbol{n}$
end if
return $\boldsymbol{g}$

Forward Pass: Active-Set Detection

Not too much theory.

1. First, use a solver `OSQP` to solve the optimization problem $\boldsymbol{z}^*(\hat{\boldsymbol{c}}) \in \arg\min_{\boldsymbol{z}} \phi(\boldsymbol{z}) + \hat{\boldsymbol{c}}^\top \boldsymbol{z}$ s.t. $\boldsymbol{A}\boldsymbol{z} = \boldsymbol{b}, \boldsymbol{G}\boldsymbol{z} \leq \boldsymbol{h}$, and obtain the primal-dual pair $(\boldsymbol{z}^*, \boldsymbol{y}^*)$.
2. Then, identify the active set $\mathcal{A}(\hat{\boldsymbol{c}})$ via complementary slackness: $\mathcal{A}(\hat{\boldsymbol{c}}) = \{i : (\boldsymbol{G}\boldsymbol{z}^*(\hat{\boldsymbol{c}}))_i = h_i\}$.
3. Finally, form the active constraint Jacobian $\boldsymbol{J} = [\boldsymbol{A}^\top, \boldsymbol{G}_{\mathcal{A}}^\top]^\top$.

Backward Pass: Efficient Gradient Computation I

Recall, given a perterbation in predicted cost $d\hat{\boldsymbol{c}}$, the corresponding perturbation in optimal solution is $d\boldsymbol{z} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}}$. Now, specify the perturbation value as the prediction error

$$d\hat{\boldsymbol{c}} \leftarrow -\boldsymbol{e}. \tag{22}$$

Plug in this specific perturbation into the sensitivity KKT system (equation 9) gives:

$$\begin{bmatrix} \boldsymbol{H} & \boldsymbol{J}^\top \\ \boldsymbol{J} & 0 \end{bmatrix} \begin{bmatrix} d\boldsymbol{z} \\ d\boldsymbol{y} \end{bmatrix} = \begin{bmatrix} \boldsymbol{e} \\ 0 \end{bmatrix} \tag{23}$$

Computing equation 23 gives:

$$d\boldsymbol{z} = \underbrace{\boldsymbol{H}^{-1}\boldsymbol{e}}_{\text{free movement}} - \underbrace{\boldsymbol{H}^{-1}\boldsymbol{J}^\top d\boldsymbol{y}}_{\text{constraint correction}}, \qquad \underbrace{\boldsymbol{J} d\boldsymbol{z}}_{\text{active constraint}} = 0. \tag{24}$$

Solving this linear system gives the projected prediction error $d\boldsymbol{z} = \boldsymbol{P}_H \boldsymbol{e}$.

Backward Pass: Efficient Gradient Computation II

It solves with some tricks as follows. Combining the two gives the reduced system:

$$\boxed{\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^{\top}d\boldsymbol{y} = \boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{e}} \tag{25}$$

To simplify notation, $\boldsymbol{x} := \boldsymbol{H}^{-1}\boldsymbol{e} \in \mathbb{R}^n$, $\boldsymbol{r} := \boldsymbol{J}\boldsymbol{x} \in \mathbb{R}^k$, and $\boldsymbol{v} := d\boldsymbol{y} \in \mathbb{R}^k$. Then the reduced system (only involves k variables) is

$$\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^{\top}\boldsymbol{v} = \boldsymbol{r} \tag{26}$$

Then once $\boldsymbol{v} = d\boldsymbol{y}$ is solved, recall that $\boldsymbol{g} = \boldsymbol{P}_H\boldsymbol{e}$ (equation 17), $d\boldsymbol{z} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}}$ (equation 11), and $d\hat{\boldsymbol{c}} = -\boldsymbol{e}$ (equation 22). These three equations give $\boldsymbol{g} = d\boldsymbol{z}$. Further by $d\boldsymbol{z} = \boldsymbol{x} - \boldsymbol{H}^{-1}\boldsymbol{J}^{\top}\boldsymbol{v}$ (equation 24), it gives

$$\boxed{\boldsymbol{g} = d\boldsymbol{z} = \boldsymbol{x} - \boldsymbol{H}^{-1}\boldsymbol{J}^{\top}\boldsymbol{v}} \tag{27}$$

References

Lee, J., Jin, S., and Lee, Y. (2026). Decision-focused learning via tangent-space projection of prediction error.